

Research Paper Comparison

Five studies benchmarked against our Smart Construction Site Safety Monitor — three original + two additional 2025 papers

Research Paper Name	Year	Model / Technique	Dataset	Main Detection	Result	Limitation / Research Gap
A YOLO-based intelligent detection algorithm for risk assessment of construction sites — Ruiyang Feng, Yu Miao, Junxing Zheng	2024	YOLOv8s	Roboflow dataset, 103,500 annotated images	PPE detection + hazardous-area entry	mAP50: 84.0%, Precision: 85.0%, Recall: 60.5% across 9 classes	Focuses mainly on vision-based detection; environmental and wearable sensing are not integrated
A real-time system for detecting personal protective equipment compliance using deep learning model YOLOv5	2024	YOLOv5	PPE image dataset	Helmet, goggles, vest, shoes, gloves, respirator, earplugs	91.5% mAP; 98.3% accuracy	Mainly focuses on PPE compliance through computer vision
Automated non-PPE detection on construction sites using YOLOv10 and transformer architectures for surveillance and body worn cameras with benchmark datasets	2025	YOLOv10 + Transformer	Benchmark datasets	Construction-site safety / non-PPE detection	Focuses on accuracy and computational efficiency	Mainly vision-based surveillance; does not combine wearable environmental sensors
SD-YOLOv5: A rapid detection method for personal protective equipment on construction sites — Li, Wang, Luo, Yin, Liu, Lu (Frontiers in Built Environment)	2025	SD-YOLOv5s (YOLOv5s + DilateFormer attention)	CHV dataset (real construction-site images)	Small-object PPE detection (helmets, vests) in complex scenes	AP: 93.7% (+2.8 pts over baseline YOLOv5s); 14.6% fewer parameters	Vision-only; no environmental or wearable sensing; needs validation on larger, more diverse datasets
Personal Protective Equipment (PPE) Monitoring for Construction Site Safety using YOLOv12 — El-Kafrawy et al. (IEEE)	2025	YOLOv12 (medium) vs. older YOLO versions	2,047 annotated images; dual T4 GPU (Kaggle)	Real-time PPE violation detection on live onsite video	Evaluated via confusion matrix, F1-score & PR curve; YOLOv12 outperformed older YOLO versions on real onsite footage	Vision-only surveillance; no wearable/environmental sensing; validated on a single real-world site

■ Highlighted rows (last two) are the newly added 2025 papers.

Key takeaway for the jury: All five studies — including both 2025 additions — remain **vision-only** systems (camera + YOLO variant). None integrate wearable IoT sensing such as fall detection, gas monitoring, or temperature tracking. This is the exact gap our Smart Construction Site Safety Monitor addresses by combining YOLO11-based detection with an ESP32 wearable smart helmet.